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Subject = Cloud Computing

Q=1⇒ Explain the architecture of a Helm Chart, including its main components.

Ans⇒ A Helm chart is a package that contains all the necessary files required to deploy an applications on Kubernetes. It follows a predefined directory structure.

Main Components of Helm chart

1. Chart.yaml/ :- This file contains metadata about the chart such as name, version, description, and dependencies.
2. Values.yaml/ :- It stores default configurations values that can be overridden during deployment.
3. templates/ :- This directory contains Kubernetes resource definitions written as templates using the Go templating engine.
4. Charts/ :- It holds dependent charts required by the main chart.
5. _helpers.tpl :- Contains reusable template functions to avoid repetition.
6. NOTES.txt :- Displays instructions to the user after installations.

Working :-

When a user runs a Helm command, Helm combines templates with values from values.yaml and generates Kubernetes manifests, which are then deployed to the cluster.

Q=2⇒ Discuss the advantages and limitations of Serverless (Faa

Ans⇒ Serverless computing allows developers to run code without managing servers.

Advantages

1. No server management :- The cloud provider manages infrastruct
2. Auto Scaling :- Applications automatically scale based on demand.
3. Cost Efficiency :- User pay only for the execution time.
4. Faster development :- Developers focus only on writing code.
5. High Availability :- Built-in redundancy and fault tolera

Limitations:

1. ~~Cloud~~ Cold Start Problem: Delay in Execution when a function is invoked after inactivity.
2. Execution Time Limits: Functions cannot run indefinitely.
3. Vendor Lock-in: Difficult to migrate between cloud providers.
4. Debugging Complexity: Hard to monitor and debug distributed systems.
5. Stateless Nature: No persistent memory between executions.

Q=3 ⇒ Explain how Google BigQuery handles large-scale analytics efficiently.

Ans ⇒ Google BigQuery is a serverless data warehouse designed for large-scale data analysis.

Key features:-

1. Columnar Storage: Data is stored in columns, reducing the amount of data scanned during queries.
2. Dremel Execution Engine: Uses a distributed architecture to process queries in parallel.
3. Serverless Architecture: Automatically manages infrastructure & scaling.
4. Separation of storage and compute: Storage and processing are handled independently for better scalability.
5. Massively Parallel Processing: Queries are executed across multiple nodes simultaneously.
6. Query Optimization: Techniques like partitioning, caching, and pruning improve performance.

Q=4 ⇒ Compare Pre-built ML APIs, AutoML, Custom ML models.

Ans ⇒ 1. Pre-built ML APIs

These are ready-to-use APIs provided by cloud platforms.

- Require minimal expertise
- Limited customization
- Suitable for common tasks like image recognition or speech

2. AutoML
AutoML tools allow users to train models with minimal coding
- Moderate level of control
- Requires labeled datasets
- Suitable for users with basic ML knowledge

3. Custom ML Models
These are fully developed models built from scratch.
- High flexibility and control
- Requires strong expertise
- Suitable for complex and specific problems.

Q=5⇒ Explain the complete lifecycle of serverless function, including trigger, execution, and scaling.
Ans⇒ The lifecycle of a serverless function consists of several stag

1. Trigger:- The function is triggered by an event such as an HTT request, file upload, or database update.
2. Initialization (Cold Start):- The system prepares the runtime environment and loads the function codes.
3. Execution:- The function processes the request and performs the required operations.
4. Scaling:- Multiple instances of the function are created automatically based on incoming requests.
5. Termination:- After execution, the function instance is either terminated or kept idle for reuse.
6. Monitoring and Logging:- Logs ~~are~~ and metrics are recorded for debugging and performance analysis.

Q=6 ⇒ Discuss the role of Helm in DevOps and CI/CD pipelin
Ans⇒ Helm plays a significant role in modern DevOps practice
Key Roles:-
1. Application Packaging:- Helm packages Kubernetes applications into charts for easy deployment.
2. Consistency Across Environments:- The same chart can be used in development, testing, and production environment.
3. Version Control and Rollbacks:- Helm maintains release history and allows easy rollback to previous versions.
4. CI/CD Integration:- Helm can be integrated with tools like Jenkins and GitHub Actions to automate deployments.
5. Configuration Management:- Values can be customized using values.yaml

Q=7 ⇒ Explain the working of edge inference with a step-by-step process.

Ans ⇒ Edge inference refers to running machine learning models directly on edge devices instead of cloud servers.

Steps Involved

1. Data Collection :- Data is collected from sensors, cameras, or IoT devices.

2. Data Preprocessing :- The collected data is cleaned and formatted

3. Model Deployment :- A pre-trained model is deployed on the edge device.

4. Inference Execution :- The model processes input data locally to generate predictions.

5. Decision Making :- Based on predictions, decisions are made in real time.

6. Action Execution :- The system performs actions such as alerts or automation.

7. Cloud Synchronization (Optional) :- Important data may be sent to the cloud for storage or further analysis.